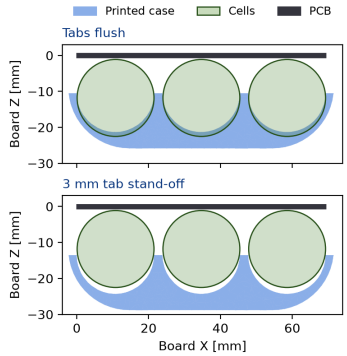
A flush case intersects all three cells before tightening



Flush tabs
The printed case overlaps each cell by about 1.18 mm at its crown. The fit develops contact load during installation.

3 mm stand-off
The unstrained STEP clears the cells and all existing mechanical hardware. Rigid sideways travel of 0.15–0.33 mm closes the first cell gap.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

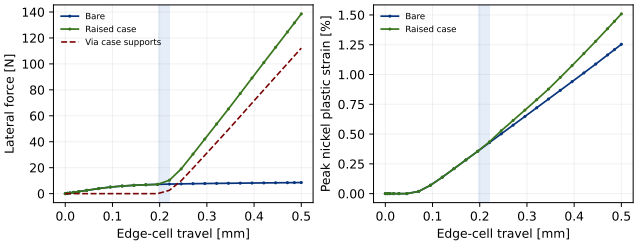
The bare-board preload repeats the October baseline

Bare-board quantity	Value	Loading
First prestressed natural frequency	485.964 Hz	Six bolts locked
Maximum PCB directional displacement	0.052674 mm RMS	Base Z
Maximum PCB equivalent stress	90.325 MPa RMS	Base X

The fresh six-bolt solve held 749.997–749.999 N at each locked fastener and reproduced the October peak static displacement of 0.04746 mm. The quantities in the table are the October dynamic reference.

The fresh bare edge-cell fixture took 8.54 N to reach 0.50 mm sideways travel. The raised case with nylon screws is the first cased comparison.

The raised nylon-screw case restrains the edge cell



The blue band brackets first contact at 0.196–0.221 mm of travel.

At the same 8.54 N pull
Cell travel fell from 0.50 mm bare to about 0.21 mm with the case.

At the same 0.50 mm travel
The pull rose from 8.54 to 138.53 N. The four case screw supports carried 112.15 N. Peak nickel plastic strain rose from 1.25% to 1.51%.

Local fixture, 3 mm stand-off, positive X. Fixed screw tops; flexible-board vibration comparison pending.